Mathematics A

Unit 1 • Chapter OL-T43 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 7 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T43 • Expertise Q16+ • 2 items • 7 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Area & Perimeter

Unit 1 • Expertise • 7 marks • 2 item(s)

January 2021 | 4MA1-2H Q18 | 3 marks

Find perimeter of a simple triangle or rectangle

January 2021 | 4MA1-2H Q20 | 4 marks

Find composite or shaded polygonal area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$
B has coordinates $(1, 10)$
C has coordinates $(19, a)$
D has coordinates (b, c)

(a) Work out the value of a, the value of b and the value of c.

$a =$ ....

$b =$ ....

$c =$ ....
(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

.... cm
(3)

Worked solution

January 2021 | Question 18 | 3 marks

Family: Find perimeter of polygonal shapes • Variant: Find perimeter of a simple triangle or rectangle

Method / working

1. AB=sqrt(5^2+12^2)=13 cm.
2. BC=sqrt(18^2+7.5^2)=19.5 cm.
3. Perimeter=2(13+19.5)=65 cm.

Final answer: 65 cm

20

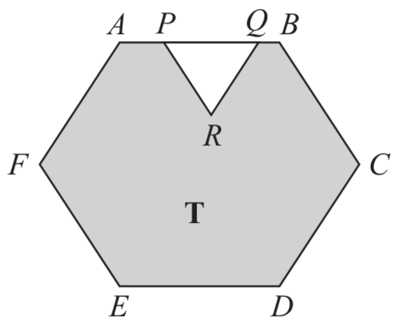

Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm²

work out the length of PQ.

.... cm

Worked solution

January 2021 | Question 20 | 4 marks

Family: Use rectangle and triangle area • Variant: Find composite or shaded polygonal area

Method / working

1. Let $PQ=x$, so $AB=1.5x$.
2. Write the equilateral triangle and regular-hexagon areas in terms of x.
3. Set shaded area equal to $72\sqrt{3}$ and solve the resulting quadratic in x^2.
4. The positive length is $PQ=4.8$ cm.

Final answer: 4.8 cm